SQL for Beginners

Beginner Study Notes and Practice Guide

This short guide introduces core concepts and provides practice ideas. It is an original learning summary, not a substitute for a complete textbook.

Introduction

SQL is used to define, query, and modify data in relational database systems.

Tables and rows

A table organizes data into columns and rows. A primary key uniquely identifies each row.

Reading data

SELECT chooses columns; WHERE filters rows; ORDER BY sorts results; LIMIT restricts the number returned in systems that support it.

Adding and changing data

INSERT adds rows, UPDATE modifies matching rows, and DELETE removes matching rows. Always check the WHERE clause before updating or deleting.

Grouping

GROUP BY combines rows into groups, often with COUNT, SUM, AVG, MIN, or MAX.

Joining tables

JOIN combines related rows from multiple tables using matching columns.

Practice

Create a Books table and query books priced below 500, then count books by category.

Practice checklist

- Review the concepts and write a small example.
- Run your example and test an edge case.
- Explain the solution in your own words.